



Internship at Bharat Space Education Research Centre

“MissionDoc AI: An Intelligent Retrieval-Augmented Document Analysis System for Satellite Missions ”

Submitted by:

SHREYAS R

PES2UG23CS564

Under the guidance of

Karan

Mentor,

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<6 weeks>

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

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DEF- SPACE SUMMER INTERNSHIP OFFER LETTER BSERC-02183

To,

Shreyas R

From

PES UNIVERSITY

We are pleased to inform you that you have been recommended for the Def-Space Summer Internship 2026 at Bharat Space Education Research Centre (भारत अंतरिक्ष शिक्षा अनुसंधान केंद्र), based on your application.

Recommended application for the Online Def-Space Summer Internship 2026 at Bharat Space Education Research Centre: We are pleased to offer you a position as Def-Space Tech Intern under the Online Def-Space Summer Internship Programme of Bharat Space Education Research Centre. You are requested to report to the undersigned at BSERC via online meeting at 12:05 PM on 19 June 2026 for the orientation session. The internship will commence on 19 June 2026 and conclude on 30 July 2026. It will be an unpaid internship with focused technical training aimed at promoting technical education at the grassroots level.

Internship Sessions & Projects (Timeline: 19th June – 30th July 2026): Def-Space Summer Internship Training Module comprises the following: (1) Generative Artificial Intelligence (AI), (2) Cyber Security & Digital Forensics in Space, Defence, UAV & AI Ecosystems including, (3) Advanced Drone Technology (Air Taxi & Defence Drone), (4) Aircraft Design Technology, (5) Rocketry Design Technology, (6) EV Technology and Charging Infrastructure for Future Engineers, (7) Robotics Design Technology, and (8) AI-Based Military Intelligence and Threat Analysis Dashboard using Machine Learning and Data Analytics.

We look forward to your valuable contribution to the online Def-Space Summer Internship Programme 2026.

Encl: As Above


Authorised Signatory
Bharat Space Education Research Centre
(भारत अंतरिक्ष शिक्षा अनुसंधान केंद्र)


निदेशक / Director
भारत अंतरिक्ष शिक्षा अनुसंधान केंद्र
Bharat Space Education Research Centre
(भारत अंतरिक्ष शिक्षा अनुसंधान केंद्र)

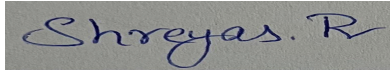
Vision of Viksit Bharat Abhiyan @2047 | Atmanirbhar Bharat (Self-Reliant India)

<https://www.bserc.org/>

DECLARATION

We hereby declare that the project entitled "**MissionDoc AI: An Intelligent Retrieval-Augmented Document Analysis System for Satellite Missions** " has been carried out at Bharat Space Education Research Centre by Shreyas R under the guidance of Karan, Mentor, and submitted in partial fulfillment of the credits for the degree of **Bachelor of Technology in Computer Science and Engineering of PES University, Bengaluru** during the academic semester 6 weeks. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

PES2UG23CS564 SHREYAS R

A rectangular box containing a handwritten signature in blue ink that reads "Shreyas. R".

ACKNOWLEDGEMENT

I would like to express my gratitude to our guide Karan, Mentor, Bharat Space Education Research Centre, for his continuous guidance, assistance and encouragement throughout the development of this project.

I am grateful to the internship coordinators Dr. Animesh Giri & Dr. Prema R, Dept. of Computer Science and Engineering, PES University – Electronic City Campus, for organizing, managing and helping out with the entire process.

I am grateful to the Capstone Project guide Dr. Kavitha, Dept. of Computer Science and Engineering, PES University – Electronic City Campus, for organizing, managing and helping out with the entire process.

I take this opportunity to thank Dr. Sandesh B J, Chairperson, Department of Computer Science and Engineering, PES University – Electronic City Campus, for all the knowledge and support I have received from the department.

I would like to thank Dr. B.K. Keshavan, Dean of Faculty, PES University, for his help.

I am deeply grateful to Prof. Jawahar Doreswamy, Chancellor, PES University, Dr. Suryaprasad J, Vice-Chancellor, PES University, and Prof. Nagarjuna Sadineni, Pro-Vice-Chancellor, PES University, for providing me various opportunities and enlightenment every step of the way.

Finally, this internship could not have been completed without the continual support and encouragement I have received from my parents and my friends.

ABSTRACT

This report presents an account of the work undertaken during the Def-Space Summer Internship 2026, an online technical training programme conducted by the Bharat Space Education Research Centre (BSERC) in association with PES University, Bengaluru, under offer reference BSERC-02183. The internship, spanning 19th June 2026 to 30th July 2026, exposed the intern to eight technology domains including Generative Artificial Intelligence, Cyber Security and Digital Forensics, Advanced Drone Technology, Aircraft Design, Rocketry Design, EV Technology, Robotics Design and AI-Based Military Intelligence and Threat Analysis.

As the primary deliverable under the Generative Artificial Intelligence track, the intern conceptualised, designed and developed the **MissionDoc AI: An Intelligent Retrieval-Augmented Document Analysis System for Satellite Missions**, a Retrieval-Augmented Generation (RAG) application built using Python, Streamlit, LangChain, PyMuPDF, ChromaDB and the Google Gemini API. The application allows a user to upload satellite-mission-related documents — ISRO annual reports, launch vehicle manuals, mission reports and research papers — and ask natural-language questions about their contents. The system extracts and chunks the text, generates vector embeddings, stores them in a persistent ChromaDB vector store, retrieves the most relevant passages through semantic similarity search, and uses the Gemini large language model to generate a grounded, source-cited answer, eliminating the need to manually search through hundreds of pages of technical documentation.

This report documents the organisational context of the internship, the responsibilities undertaken, the technical accomplishments achieved in building the assistant, the challenges encountered during development, and the key lessons learned. It concludes with reflections on the overall value of the internship and recommendations for future participants and programme organisers.

I. INTRODUCTION

The Bharat Space Education Research Centre (BSERC) is an organisation working under the broader vision of "Viksit Bharat Abhiyan @2047" and "Atmanirbhar Bharat" (Self-Reliant India), with a mission to promote technical education at the grassroots level in the fields of defence and space technology. Operating in the Def-Space Education and Innovation space, BSERC designs structured training programmes intended to give students hands-on exposure to emerging technologies that support India's aerospace and defence innovation ecosystem.

The Def-Space Summer Internship 2026 was conducted as an online programme in collaboration with PES University, Bengaluru. The internship was offered through a selection test, following which the intern, Shreyas R, a student of PES University, Bengaluru, was offered the position of Def-Space Tech Intern under offer reference BSERC-02183. The intern was required to report for an orientation session conducted online at 12:05 PM on 19th June 2026, following which the internship formally commenced. It was structured as an unpaid internship focused on technical training, running from 19th June 2026 to 30th July 2026.

The programme was structured around six broad tech domains and eight distinct project tracks, namely Generative Artificial Intelligence, Cyber Security and Digital Forensics in Space, Defence, UAV and AI Ecosystems, Advanced Drone Technology (Air Taxi and Defence Drone), Aircraft Design Technology, Rocketry Design Technology, EV Technology and Charging Infrastructure for Future Engineers, Robotics Design Technology, and AI-Based Military Intelligence and Threat Analysis Dashboard using Machine Learning and Data Analytics.

Within this structure, the intern was placed under the Generative Artificial Intelligence track. This track focused on applying large language models and retrieval-based AI techniques to problems relevant to the space and defence domain, rather than treating generative AI as an isolated academic exercise. The intern's role fit within this track as an individual contributor responsible for the end-to-end design, development and testing of a document-intelligence application aimed at a genuine pain point in the space sector:

the difficulty of locating specific information buried inside long, dense technical documents such as ISRO annual reports and launch vehicle manuals. The intern worked independently on the implementation while following the technical milestones and periodic mentorship guidance provided by BSERC.

II. JOB RESPONSIBILITIES AND DUTIES

As a Def-Space Tech Intern on the Generative Artificial Intelligence track, the intern was assigned the responsibility of designing and building a functional application that demonstrated a practical, non-trivial application of large language models to a real information-retrieval problem in the space and defence documentation domain. This led to the selection of Retrieval-Augmented Generation (RAG) as the core technique, since it directly addresses the two weaknesses of raw LLM usage that are most damaging in a technical/defence context: hallucination and the inability to reference an organisation's own private documents.

Core responsibilities:

- Researching the RAG paradigm and evaluating how it could be applied to satellite mission documentation, launch vehicle manuals and ISRO annual reports.
- Designing the end-to-end system architecture, covering document ingestion, chunking, embedding generation, vector storage, retrieval and answer generation.
- Selecting and integrating the technology stack: Python as the core language, Streamlit for the user interface, PyMuPDF for PDF text extraction, LangChain for text splitting and orchestration, ChromaDB as the persistent vector database, and the Google Gemini API as the generation model.
- Implementing the PDF ingestion pipeline, including text extraction from multi-page technical PDFs and handling of formatting inconsistencies commonly found in scanned or exported government/organisational reports.
- Implementing a recursive character-based chunking strategy to split extracted text into overlapping segments suitable for embedding, tuned so that individual chunks remained semantically coherent without losing surrounding context.
- Generating vector embeddings for each chunk and persisting them in ChromaDB, along with metadata (source filename and page number) required for citation.
- Building the semantic retrieval component, which converts a user's question into

a query embedding and performs a similarity search against the vector store to fetch the most relevant chunks.

- Designing prompt templates that instruct the Gemini model to answer strictly from the retrieved context and to decline or hedge when the retrieved evidence is insufficient, reducing hallucinated answers.
- Implementing a similarity-score-based confidence indicator, shown to the user alongside each answer, so that low-confidence retrievals are visibly flagged rather than presented with false certainty.
- Building the Streamlit front end, including a sidebar for API key configuration and document upload/processing, a chat-style question-answering interface, an expandable "Sources" panel showing the exact source document and page number for each citation, and live statistics on the number of documents and chunks currently indexed in the knowledge base.
- Testing the application against multiple real ISRO annual reports and mission documents to validate retrieval accuracy, citation correctness and response quality.

The development process followed an iterative, milestone-driven methodology aligned with the six-week structure of the internship. In the initial phase, the intern focused on requirement analysis and architecture design, finalising the choice of ChromaDB over alternative vector stores after evaluating ease of local persistence, and finalising Gemini as the generation model after evaluating its context-window size and citation-friendliness against other available APIs. The middle phase was dedicated to core development: building the ingestion and chunking pipeline, wiring up the embedding and retrieval logic, and getting a working end-to-end query flow. The final phase focused on prompt refinement to reduce hallucination, tuning chunk size and overlap to improve retrieval quality, UI polishing, and testing the assistant against real ISRO Annual Report PDFs to confirm that answers were both accurate and properly cited.

The intern's work contributed directly to the broader goal of the Generative AI track, which was to give participants hands-on, deployable experience with real-world

applications of large language model APIs rather than purely theoretical exposure. The completed assistant, tested end-to-end against genuine ISRO documentation, stands as a concrete milestone completed within the internship's project timeline of 19th June to 30th July 2026.

Since the internship was conducted entirely online, collaboration with the mentor took place through virtual meetings and asynchronous communication. Periodic check-ins with BSERC's technical mentors provided guidance on retrieval-quality issues, prompt design for grounded answers, and general software architecture decisions, while also creating opportunities to discuss challenges encountered during development and receive feedback on the evolving project.

III. ACCOMPLISHMENTS AND ACHIEVEMENTS

The most significant achievement of the internship was the successful design, development and testing of the **MissionDoc AI: An Intelligent Retrieval-Augmented Document Analysis System for Satellite Missions** , a working RAG application that allows a user to query large satellite-mission PDFs in natural language and receive grounded, cited answers. The completed project demonstrates practical competence in document processing, embedding-based retrieval, vector database usage, prompt engineering and applied LLM integration, applied to a genuine use case rather than a toy example.

Key features successfully implemented:

- PDF upload and automated text extraction using PyMuPDF, supporting multi-page ISRO annual reports and mission documents.
- Recursive text chunking to split extracted content into overlapping, semantically coherent segments suitable for embedding.
- Vector embedding generation for every chunk, with persistent storage in ChromaDB so the knowledge base survives across sessions.
- Semantic similarity search to retrieve the most relevant chunks for a given user question.
- Grounded question-answering using the Google Gemini API, constrained by prompt design to answer from retrieved context.
- Source citation for every generated answer, displaying the exact source document and page number, with a "Sources" panel for full transparency.
- A similarity-score-based confidence indicator (for example, flagging an answer as "High confidence" when the top retrieval match exceeds a set similarity threshold), giving the user a signal for how much to trust a given response.
- A knowledge-base statistics panel showing the live count of ingested documents and generated chunks.
- A clean, chat-style Streamlit interface with a document-ingestion sidebar, an expandable "How this works" explainer, and a persistent chat history.

The application was tested end-to-end against real ISRO Annual Report PDFs. In one representative test run, the assistant ingested a single 2023–24 ISRO Annual Report document, which was split into 494 chunks and indexed successfully. When asked a factual question about the Chandrayaan-3 mission, the assistant correctly identified the spacecraft's launch vehicle, launch date and mass, its Propulsion Module–Lander–Rover configuration, and the dates of lunar orbit insertion and lander separation — every claim in the response was traceable to a specific page of the source PDF, and the system reported a top-match similarity score of 0.71, correctly labelled as high confidence. This confirmed that the retrieval pipeline was surfacing relevant chunks and that the generation prompt was successfully constraining the model to ground its answer in the retrieved evidence rather than relying on the model's own background knowledge.

In terms of measurable outcomes, the ingestion pipeline reliably processed multi-page PDF documents without manual pre-formatting, the chunking and embedding pipeline produced hundreds of indexed chunks from a single document within a short processing time, and the retrieval-and-generation loop returned a cited, grounded answer to a natural-language query in a few seconds. These outcomes validate the core objective of the Generative AI track: producing a genuinely usable application of retrieval-augmented generation, rather than a proof-of-concept confined to a notebook.

During development, the intern encountered and overcame several technical challenges. Dependency conflicts between LangChain, ChromaDB and the Google Gemini SDK were resolved by carefully pinning package versions and isolating the project in a dedicated Python environment. Early retrieval quality was poor when chunks were too large or too small; this was addressed by iteratively tuning the chunk size and overlap parameters and observing their effect on retrieval relevance. Initial responses occasionally drifted from the source material; this was resolved through prompt engineering, explicitly instructing the model to restrict itself to the retrieved context and to state when the answer could not be found in the documents. Extracting clean text from PDFs with complex layouts, tables and embedded images required additional

handling in the PyMuPDF extraction logic to avoid garbled or missing text.

As evidence of these accomplishments, the working application, its ingestion of a genuine ISRO Annual Report into a persistent ChromaDB knowledge base, and its correctly cited, high-confidence responses to test queries serve as verifiable proof of the project's completion and functionality.

IV. LESSONS LEARNED

The Def-Space Summer Internship 2026 provided valuable, practical insight into applying generative AI to a genuine information-retrieval problem rather than a purely theoretical setting. Building the AI Satellite Mission Document Assistant from the ground up reinforced the intern's understanding of how the individual stages of a RAG pipeline — extraction, chunking, embedding, retrieval and generation — interact with one another, and how each stage's quality directly bounds the quality of the final answer.

Key lessons and insights gained during the internship include:

- A deeper practical understanding of Retrieval-Augmented Generation as an architecture, and why it is preferable to raw LLM prompting when answers must be grounded in a specific, private set of documents.
- Hands-on experience with vector embeddings and vector databases, including how ChromaDB stores and indexes embeddings for efficient similarity search.
- The importance of chunking strategy — chunk size, overlap and boundary choice have a direct and sometimes counter-intuitive effect on retrieval quality.
- Practical prompt engineering skills, particularly techniques for constraining a large language model to answer only from supplied context and to avoid hallucinating information not present in the source documents.
- The value of building in transparency features, such as source citation and confidence scoring, when deploying an LLM-based tool for use with authoritative technical documents.
- Improved skills in using LangChain to orchestrate a multi-stage AI pipeline, and in integrating a commercial LLM API (Google Gemini) into a Python application.
- Experience building an interactive Streamlit application, including sidebar-driven configuration, file upload handling and stateful chat interfaces.
- The discipline of testing an AI system against real-world documents rather than curated examples, which surfaced practical issues (messy PDF layouts, edge cases in extraction) that would not have appeared with clean synthetic test data.

The internship helped the intern move from theoretical familiarity with large language models to the practical, applied skill of building a complete retrieval-augmented system end to end. Going forward, the intern intends to build on these skills by exploring more advanced retrieval techniques such as hybrid search and re-ranking, and by applying the same disciplined, evidence-grounded approach to future generative AI projects.

In terms of areas for further development, the intern recognises the value of gaining deeper expertise in evaluation methodologies for RAG systems — such as systematic measurement of retrieval precision and answer faithfulness — which were assessed informally during this project but would benefit from a more rigorous, metrics-driven approach in future work. Additionally, further exploration of advanced chunking strategies (such as semantic or structure-aware chunking for tables and figures) would help improve retrieval quality on documents with complex, non-linear layouts.

V. CONCLUSION

The Def-Space Summer Internship 2026, conducted by the Bharat Space Education Research Centre in collaboration with PES University, Bengaluru, provided a valuable, hands-on opportunity to apply generative AI and retrieval-based system design to a real-world documentation problem in the space sector. The internship's most significant outcome was the successful development and testing of the MissionDoc AI: An Intelligent Retrieval-Augmented Document Analysis System for Satellite Missions , a Retrieval-Augmented Generation application built using Python, Streamlit, LangChain, ChromaDB and the Google Gemini API. This project demonstrated the intern's ability to design a multi-stage AI pipeline, implement grounded and cited question-answering, and validate the system against genuine ISRO documentation rather than synthetic test cases.

The intern is grateful to the Bharat Space Education Research Centre and its Director for the opportunity to participate in this internship, and to PES University, Bengaluru, for facilitating the recommendation. The guidance and mentorship provided by Karan throughout the programme were instrumental in helping the intern navigate technical challenges — particularly around retrieval quality and prompt design — and complete the project successfully.

Based on this experience, it is recommended that future iterations of the Def-Space Summer Internship continue to emphasise project-based, deployable outcomes over purely theoretical instruction, as this approach proved highly effective in building practical, domain-relevant skills. It is also recommended that future interns on the Generative AI track be given early exposure to RAG evaluation techniques and structured testing against real documents, which would further strengthen the reliability of the AI systems developed during the programme.

VI. APPENDICES

Appendix A: Internship Offer Letter

Offer Reference: BSERC-02183, issued by the Bharat Space Education Research Centre to Shreyas R (PES University, Bengaluru), confirming the offer of the position of Def-Space Tech Intern under the Def-Space Summer Internship 2026, with the internship period from 19th June 2026 to 30th July 2026.

Appendix B: Project Repository and Live Deployment

Source Code (GitHub): <https://github.com/Shreyas-9900/MissionDoc-AI>

Live Demo: <https://missiondoc-ai.onrender.com>

Appendix C: Technology Stack Summary

- **Programming Language:** Python
- **User Interface:** Streamlit
- **PDF Text Extraction:** PyMuPDF
- **Document Chunking & Orchestration:** LangChain
- **Vector Database:** ChromaDB (Persistent Storage)
- **AI Integration:** Google Gemini API (Embeddings & Large Language Model)
- **Core Technique:** Retrieval-Augmented Generation (RAG)
- **Knowledge Retrieval:** Semantic Similarity Search
- **Repository Structure:** `app.py`, `ingestion.py`, `rag_chain.py`, `requirements.txt`, `README.md`, `.env.example`

Appendix D: Application Workflow

The end-to-end workflow of the **MissionDoc AI: An Intelligent Retrieval-Augmented Document Analysis System for Satellite Missions**, from document upload to answer generation, proceeds through the following stages:

- **Document Upload:** The user uploads one or more satellite mission-related PDF documents (such as ISRO annual reports, launch vehicle manuals, mission reports, or research papers) through the Streamlit interface.
- **Text Extraction:** The application uses **PyMuPDF** to extract textual content from each uploaded PDF document while preserving page-level metadata.
- **Document Chunking:** The extracted text is divided into overlapping chunks

using **LangChain's Recursive Character Text Splitter**, enabling efficient semantic indexing and retrieval.

- **Embedding Generation:** Each text chunk is converted into a high-dimensional vector embedding using the **Google Gemini Embedding Model**.
- **Vector Storage:** The generated embeddings, along with their corresponding document names and page numbers, are stored in a persistent **ChromaDB** vector database for efficient semantic search.
- **Query Processing:** The user enters a natural-language question through the application's chat interface.
- **Semantic Retrieval:** The user's query is embedded and compared with the stored document embeddings to retrieve the most semantically relevant text chunks.
- **Answer Generation:** The retrieved context and the user's question are provided to the **Google Gemini Large Language Model**, which generates a grounded response based solely on the retrieved information.
- **Citation Display:** The final response is displayed together with the corresponding source document, page number, retrieved passages, and a similarity-based confidence indicator, enabling users to verify the generated answer.

Appendix D: Sample Query and Response

As a representative test case, the assistant was asked "*What is Chandrayaan-3*" against an ingested ISRO Annual Report 2023–24. The system returned a grounded answer describing the spacecraft's mass, launch vehicle and launch date, its Propulsion Module–Lander–Rover architecture, and key mission-timeline dates, with every claim cited to a specific page of the source document and an overall confidence rating of "High confidence (top match similarity: 0.71)".

All materials referenced in this report, including the offer letter and the application described above, are credited to their original sources: the offer letter to the Bharat Space Education Research Centre, and the ISRO Annual Report data used for testing to the Indian Space Research Organisation.

Closing Remarks

The successful completion of the **MissionDoc AI: An Intelligent Retrieval-Augmented Document Analysis System for Satellite Missions** marks the culmination of the knowledge and practical experience gained during the **Def-Space Summer Internship 2026**. The project provided valuable exposure to modern Artificial Intelligence technologies, Retrieval-Augmented Generation (RAG), Large Language Models, semantic search, vector databases, and real-world software development practices.

Beyond the technical implementation, the internship strengthened problem-solving abilities, software engineering principles, debugging skills, and collaborative learning. The experience demonstrated how Artificial Intelligence can be effectively applied to simplify access to complex technical documentation, making information retrieval more accurate, efficient, and user-friendly.

The knowledge and skills acquired throughout this internship will serve as a strong foundation for future academic pursuits, research, and professional development in the fields of Artificial Intelligence, Machine Learning, and Software Engineering.

End of Report